

3 The enzyme urease catalyses the following reaction:



The effect of pH on this reaction was investigated using the following method.

- One test tube containing 5 cm³ of urease solution was placed in a water bath at 40 °C and left for 10 minutes.
 - Another test tube containing 5 cm³ of urea solution in a buffer at pH 3.0 was placed in the same water bath and left for 10 minutes.
 - After 10 minutes, the contents of both tubes were mixed together in one test tube.
 - This test tube was replaced in the water bath.
 - The concentration of ammonia was measured after 15 minutes and again after 60 minutes.
 - The procedure was repeated for pH values of 4.0, 6.5, 6.8, 7.3, 8.0 and 9.0.
- (a) (i) Explain why the urease solution and the urea solution were kept in the water bath at 40 °C before and after being mixed.

(4)

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- (ii) Explain how you could determine suitable concentrations of urea and urease solutions to use in this investigation.

(2)

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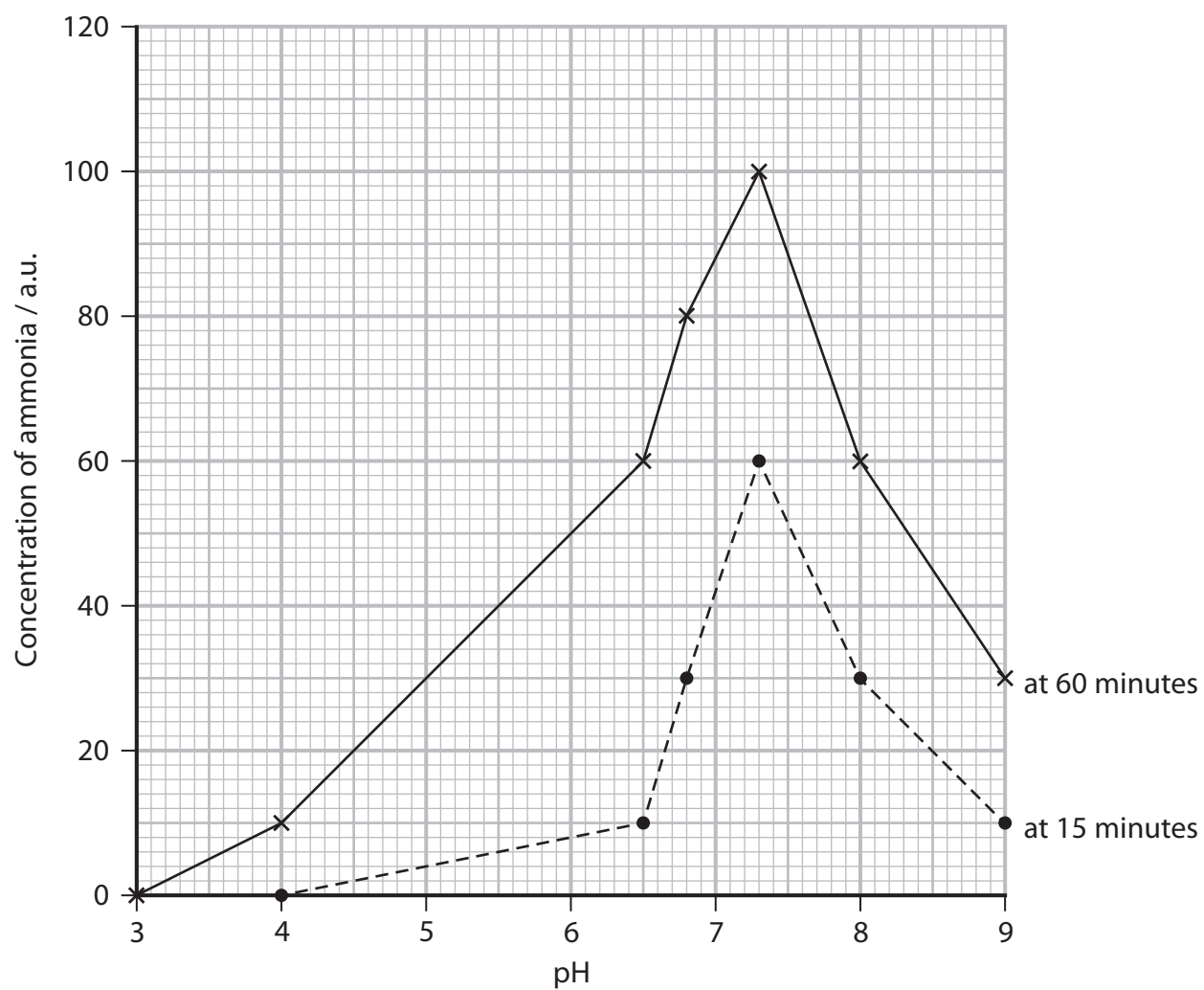
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(b) The graph shows the results of this investigation.



(i) Draw a suitable table to include the results for the experiment at 60 minutes.

(4)



- (ii) Calculate the rate of reaction at a pH of 7.3 during the first 15 minutes of this investigation.

Include appropriate units with your answer.

(2)

Answer

- (iii) Compare and contrast the results shown in the graph at 60 minutes with those at 15 minutes.

(3)

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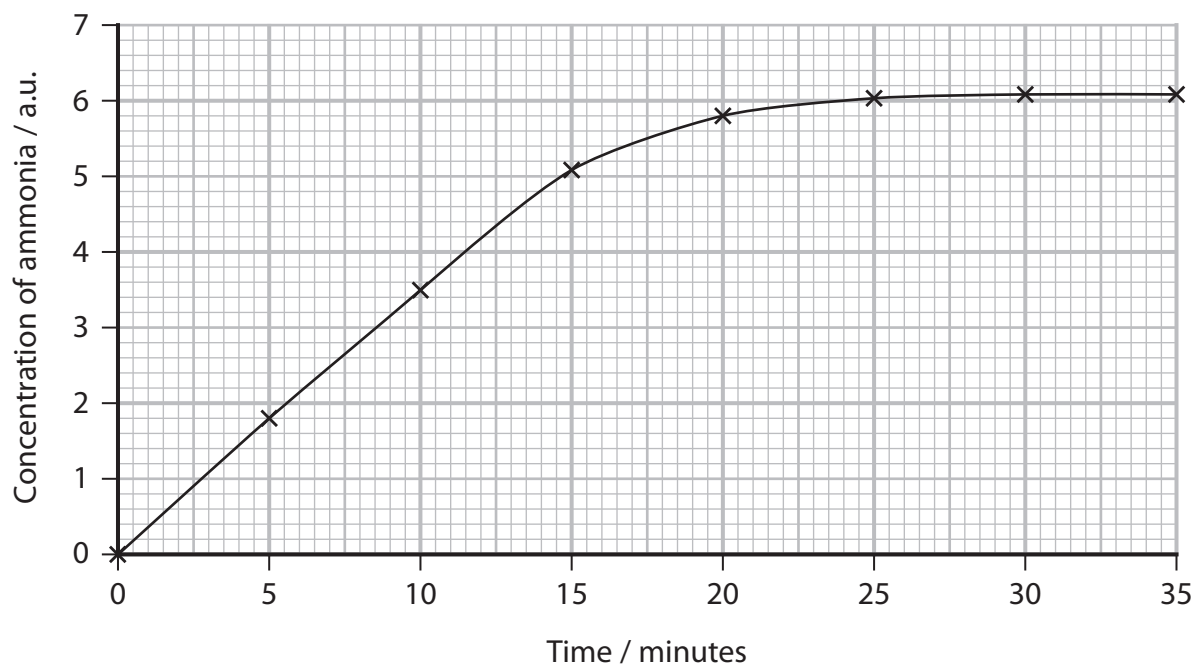
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- (c) In another investigation, the concentration of ammonia was recorded every 5 minutes for 35 minutes, at a pH of 6.0.

The graph shows the results of this investigation.



Calculate the initial rate of this reaction.

Include appropriate units with your answer.

(1)

Answer

(Total for Question 3 = 16 marks)

TOTAL FOR PAPER = 50 MARKS

